

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-1027

St. Martins 138 kV
June 20, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1027. The Interconnection Customer proposes a 123.9 MW solar photovoltaic facility interconnecting at the St. Martins 138 kV substation in the Virginia area. The request was received on July 15, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 433 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$23,444,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
S BELLEVILLE - NORTHPOINT 138 kV	PAGE - PURDY 138 kV	128.5	360	28.1
HOUSTON - ST LOUIS 138 kV L	ST LOUIS - NUCLEAR FUELS 138 kV	126.8	178	15.7
NUCLEAR FUELS - ST. MARTINS 138 kV L	WADSWORTH - ROOSEVELT RD 138 kV	126.0	193	56.3
ABBOTSFORD - BRIDGE RIV 138 kV	BROOKLYN CITY N PLANT - MISET 138 kV	122.2	222	5.4
NORTHPOINT - MITCHELL GARDENS 138 kV	PAGE - PURDY 138 kV Line	117.6	379	28.1
ST LOUIS - NUCLEAR FUELS 138 kV	WADSWORTH - ROOSEVELT RD 138 kV	113.0	329	22.1
KAPSTONE - MITCHELL GARDENS 138 kV	PURDY - PAGE 138 kV L	111.5	391	28.1
BLACKBERRY - CHATOM 138 kV L	PURDY - CHATOM 138 kV	111.3	227	11.9
VIDEO PD - BOXELDER 138 kV Line	ST LOUIS - NICHOLS 138 kV	111.0	241	8.0
BLACKBERRY SUBSTATION - NEW SALISBURY 138 kV Line	PURDY - CHATOM 138 kV	110.6	224	11.9
TIMBERVILLE - NEW SALISBURY 138 kV	PURDY - CHATOM 138 kV	109.3	223	11.9
TIMBERVILLE - FRITO LAY 138 kV	PURDY - CHATOM 138 kV	108.3	220	11.9
BROOKLYN CITY NORTH PLANT - MIDSET 138 kV line	BRIDGE RIVER - ABBOTSFORD 138 kV line	102.4	277	5.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
S BELLEVILLE - NORTHPOINT 138 kV	Rebuild line with higher-capacity conductor	\$8,217,000	20% (\$1,658,000)
HOUSTON - ST LOUIS 138 kV L	Rebuild line with higher-capacity conductor	\$4,263,000	39% (\$1,682,000)
NUCLEAR FUELS - ST. MARTINS 138 kV L	Rebuild line with higher-capacity conductor	\$5,884,000	72% (\$4,258,000)
ABBOTSFORD - BRIDGE RIVER 138 kV L	Rebuild line with higher-capacity conductor	\$10,717,000	27% (\$2,902,000)
NORTHPOINT - MITCHELL GARDENS 138 kV	Rebuild line with higher-capacity conductor	\$9,471,000	20% (\$1,912,000)
ST LOUIS - NUCLEAR FUELS 138 kV	Rebuild line with higher-capacity conductor	\$6,110,000	25% (\$1,531,000)
MITCHELL GARDENS - KAPSTONE 138 kV Line	Rebuild line with higher-capacity conductor	\$4,915,000	20% (\$992,000)
BLACKBERRY - CHATOM 138 kV	Rebuild line with higher-capacity conductor	\$6,277,000	39% (\$2,464,000)
BOXELDER - VIDEO PD 138 kV	Rebuild line with higher-capacity conductor	\$7,240,000	14% (\$1,010,000)
BLACKBERRY - NEW SALISBURY 138 kV	Rebuild line with higher-capacity conductor	\$6,616,000	39% (\$2,597,000)
TIMBERVILLE - NEW SALISBURY 138 kV	Reconductor line	\$2,287,000	39% (\$898,000)
TIMBERVILLE - FRITO LAY 138 kV	Reconductor line	\$1,800,000	39% (\$706,000)
BROOKLYN CITY NORTH PLANT - MIDSET 138 kV line	Reconductor line	\$2,737,000	30% (\$834,000)
	Total allocated to project		\$23,444,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.